

Brewster's Angle

Timothy Lim

February 6, 2026

1 Abstract

This report investigates the refractive index (n) and Brewster's angle (θ_{Br}) of a D-shaped acrylic lens through three independent methods: Snell's Law, the Polarization Method, and the Critical Angle Method. While the apparatus was commercially labeled as Acrylic ($n \approx 1.489$), experimental results consistently indicated a lower-index material. By combining data from all three methods using a weighted average, the refractive index was determined to be $\bar{n} = 1.466 \pm 0.009$, corresponding to a Brewster's angle of $\bar{\theta}_{Br} = 55.7 \pm 0.2^\circ$. These results provide a validation of Maxwell's electromagnetic theory, specifically the minimum of p-polarized reflection at the Brewster angle. Furthermore, the 0.8σ internal agreement between the polarization and critical angle methods suggests that the discrepancy with literature values is likely due to the true material of the lens rather than experimental error.

2 Introduction and Theory

In this report, 'p' and 's' as subscripts denotes calculations for light whose E field is polarised parallel and perpendicular respectively, to the plane of incidence.

The Fresnel Equations, upon simplification by using Snell's Law, gives us these 4 equations, where r and t represent the amplitude reflection coefficient and amplitude transmission coefficient respectively:

$$r_s = -\frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)}$$

$$t_s = \frac{2 \sin \theta_t \cos \theta_i}{\sin(\theta_i + \theta_t)}$$

$$r_p = \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)}$$

$$t_p = \frac{2 \sin \theta_i \cos \theta_t}{\sin(\theta_i + \theta_t) \cos(\theta_i - \theta_t)}$$

where θ_i is the angle of incidence and θ_t is the angle of refraction. By observing the equation for r_p , when $\theta_i + \theta_t = 90^\circ$, $\tan(\theta_i + \theta_t) \rightarrow \infty$, and thus $r_p \rightarrow 0$. At this specific angle (also known as Brewster's Angle), 100% of the reflected light is polarised perpendicular to the plane of incidence. This suggests a possible method to measure the refractive index of the acrylic lens: vary the angle of incidence, then measure the ratio of parallel polarised light to perpendicularly polarised light at each angle. When the ratio of parallel polarised reflected intensity to total reflected intensity approaches zero, $\theta_i = \theta_{Br}$. Starting with Snell's Law,

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

where n_1, n_2 are the refractive indices in the incident medium and refracted medium respectively, and $\theta_1,$

θ_2 are the angles of incidence and refraction respectively. When $\theta_i = \theta_{Br}$,

$$\begin{aligned}\theta_{Br} + \theta_2 &= 90^\circ \\ \theta_2 &= 90^\circ - \theta_{Br} \\ \sin \theta_2 &= \sin(90^\circ - \theta_{Br}) \\ \sin \theta_2 &= \cos \theta_{Br}\end{aligned}\tag{1}$$

Substitute equation (1) into Snell's Law to obtain:

$$\begin{aligned}n_1 \sin \theta_{Br} &= n_2 \cos \theta_{Br} \\ \tan \theta_{Br} &= \frac{n_2}{n_1}\end{aligned}$$

Finally, as the incident medium in this experiment was air, its estimated refractive index $n_1 \approx 1.00$, which results in:

$$\tan \theta_{Br} = n_2\tag{2}$$

which is the relationship between the Brewster Angle and the refractive index of the acrylic.

3 Experimental Setup

The experiment was conducted with the following setup:

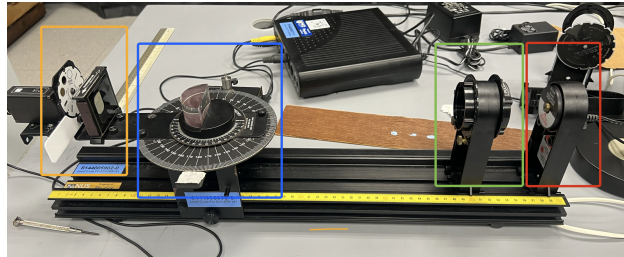


Figure 1: The experimental setup for measuring Brewster's Angle

Examining the equipment from left to right:

- **Orange:** Light sensor to measure the intensity of the polarised light, and a square polariser to allow either parallel or perpendicularly polarised light to pass through.
- **Blue:** The acrylic D-lens mounted on a goniometer.
- **Green:** Two polarisers mounted side by side: the polariser closer to the laser diode (red square) was used to control the intensity of the laser light, while the second polariser was fixed at 45° .
- **Red:** The laser diode emitting red light ($\approx 650nm$).

Before each experiment was conducted, the D-lens was calibrated by first removing the lens from the setup, setting the goniometer to 180° , then centering the laser beam on the centre of the slit in front of the sensor. The D-lens and lens holder were then placed back into the setup and the D-lens was moved laterally until the laser beam returned to the centre of the slit again.

3.1 Precautions

Due to the potential eye hazards associated with direct laser exposure, several safety protocols were strictly followed throughout the experiment.

- The "Laser In Use" sign outside the room was turned on throughout the experiment.
- All participants were required to maintain their line of sight above the horizontal plane of the laser beam to prevent accidental eye exposure. The laser was also deactivated prior to any close-range inspection of the equipment.

4 Experiment 1: Snell's Law

4.1 Procedure

This experiment is based on Snell's Law, $n_1 \sin \theta_1 = n_2 \sin \theta_2$. As the incident medium is air, approximating $n_1 \approx 1.00$ produces the equation $n_2 = \frac{\sin \theta_1}{\sin \theta_2}$, which will be used to calculate the refractive index of the acrylic lens (n_2).

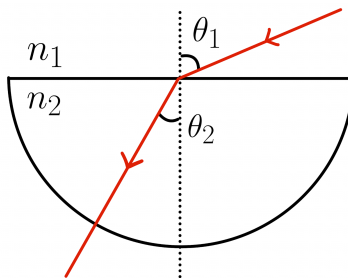


Figure 2: Incident(θ_1) and Refracted Rays(θ_2)

The independent variable for this experiment was the angle of incidence, θ_1 , and the dependent variable was the angle of refraction, θ_2 . All other variables were kept constant, such as:

- wavelength of laser light
- incident and transmission medium
- location on the D-lens where the light was incident upon

A total of 8 incident angles were studied, ranging from 10° to 80° in intervals of 10° . For each angle of incidence, 3 readings of the angle of refraction were measured, and the average of the 3 angles were taken.

4.2 Calculated Values and Error Propagation

The steps taken to obtain the refractive index and Brewster's Angle are as follows.

1. The uncertainty of θ was estimated as half of the smallest interval of the goniometer.

$$\delta\theta = 0.5^\circ = \frac{1}{180}\pi \text{ rad} \quad (3)$$

2. The uncertainties of $\sin \theta$ for each value of θ_1 and θ_2 were calculated using¹:

$$\sigma_{\sin \theta} = |\cos \theta| \times \delta\theta \quad (4)$$

¹J.R. Taylor. *An Introduction to Error Analysis* (2nd Edition), Eq. 3.23

3. The uncertainties for each value of $\frac{\sin \theta_1}{\sin \theta_2}$ were calculated using the uncertainties of $\sin \theta_1$ and $\sin \theta_2$ added in quadrature².

$$\sigma = \delta\left(\frac{\sin \theta_1}{\sin \theta_2}\right) = \left|\frac{\sin \theta_1}{\sin \theta_2}\right| \times \sqrt{\left(\frac{\delta \sin \theta_1}{\sin \theta_1}\right)^2 + \left(\frac{\delta \sin \theta_2}{\sin \theta_2}\right)^2} \quad (5)$$

4. Due to the high fractional uncertainties at smaller angles, there was a large range of uncertainties (max uncertainty $\approx 10 \times$ min uncertainty). As such, a linear weighted least squares fit was chosen. The weights were given by³

$$w_i = \frac{1}{\sigma_i^2} \quad (6)$$

Two parameters were used to fit the linear graph, one for the gradient(m) and one for the vertical intercept(C). The Excel Solver tool was used to minimise the sum of squared residuals by varying these parameters.

5. After the parameters were obtained, the errors were calculated⁴:

$$\sigma_m = \sqrt{\frac{\sum wx^2}{\Delta}} \quad (7)$$

$$\sigma_c = \sqrt{\frac{\sum w}{\Delta}} \quad (8)$$

where Δ is defined as⁵

$$\Delta = \sum w \sum wx^2 - (\sum wx)^2 \quad (9)$$

6. The refractive index and its uncertainty were then calculated:

$$n_2 = 1/m \quad (10)$$

$$\sigma_n = \left|\frac{1}{m^2}\right| \times \sigma_m \quad (11)$$

7. Finally, Brewster's Angle and its uncertainty were calculated:

$$\theta_{Br} = \tan^{-1}(n_2) \quad (12)$$

$$\sigma_{\theta_{Br}} = \left|\frac{1}{1+n^2}\right| \times \sigma_n \quad (13)$$

²J.R, Taylor. *An Introduction to Error Analysis* (2nd Edition), Eq. 3.18

³Ibid, Eq. 7.11

⁴Ibid, Problem 8.19

⁵Ibid, Eq. 8.39

4.3 Results

The experimental values were plotted in the graph below.

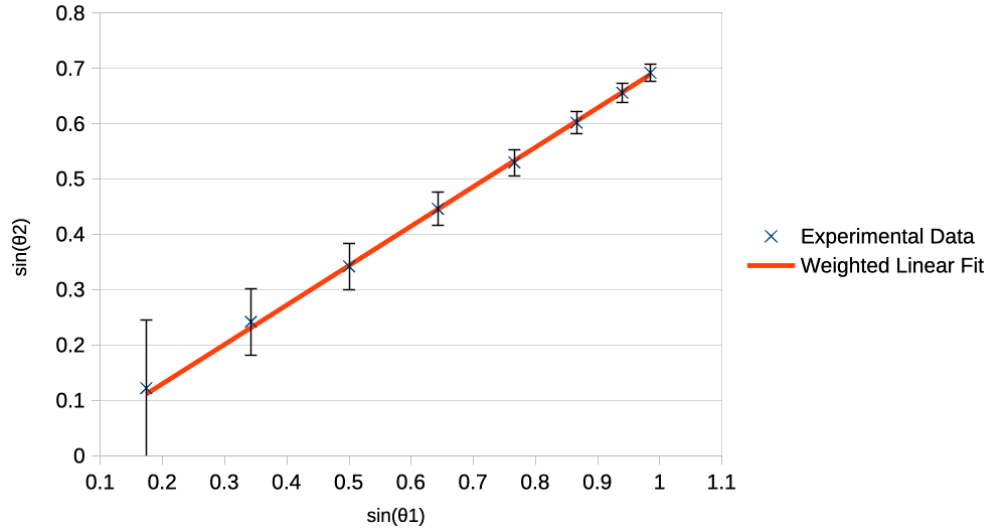


Figure 3: Graph of $\sin \theta_2$ against $\sin \theta_1$. Error bars are calculated in step 2 of the previous section.

The parameters of the least squares fit were

$$m = 0.71 \pm 0.05 \quad (14)$$

and

$$C = -0.01 \pm 0.05 \quad (15)$$

The refractive index was calculated to be

$$n_2 = 1.40 \pm 0.09 \quad (16)$$

which corresponds to a Brewster's Angle of

$$\theta_{Br} = 55 \pm 2^\circ \quad (17)$$

The R^2 value for this fit is $R^2 = 0.999$. The high coefficient of determination validates the precision of the measured refractive index. This value establishes a reference point for calculating the expected Brewster angle in subsequent experiments.

A detailed comparison with accepted values and analysis of systematic errors is presented in Section 8: Discussion.

5 Experiment 2: Polarization Method

5.1 Procedure

Following the calibration of the D-lens, the lens assembly was rotated to an incident angle of 85° , and the sensor arm was aligned to the optical path. Adjustments were made to the rotating polarizer to ensure the sensor signals remained below saturation levels ($< 100\%$).

For each angular position, the angle of maximum intensity (θ_1) was identified. The square polarizer was then inserted to measure the intensity of parallel polarized light (R_p), followed by a 90° rotation to measure the perpendicular polarized intensity (R_s). Measurements were conducted from 85° to 25° , in intervals of 5° . To accurately resolve the minimum point of the graph, the sampling interval was decreased from 5° to 1° as the incident angle approached the theoretical Brewster angle. The incident angle θ_1 was calculated from x , the value displayed on the computer by the software connected to the goniometer, using the relationship:

$$\theta_1 = \frac{1}{2} \left[180^\circ - \frac{1}{15} |x|^\circ \right] \quad (18)$$

5.2 Calculated Values and Error Propagation

The method used to obtain the refractive index, Brewster's Angle, and their associated uncertainties are as follows:

1. The theoretical framework is established using the Fresnel equations for the amplitude reflection coefficients of the perpendicular (r_s) and parallel (r_p) components:

$$r_s = \frac{n_1 \cos \theta_1 - n_2 \cos \theta_2}{n_1 \cos \theta_1 + n_2 \cos \theta_2} \quad (19)$$

$$r_p = \frac{n_2 \cos \theta_1 - n_1 \cos \theta_2}{n_2 \cos \theta_1 + n_1 \cos \theta_2} \quad (20)$$

The intensities are obtained by squaring the amplitude coefficients ($R_s = r_s^2$ and $R_p = r_p^2$). The experimental observable is the ratio of the parallel reflected intensity to the total reflected intensity.

$$\frac{R_p}{R_p + R_s} = \frac{\cos^2(\theta_1 + \theta_2)}{\cos^2(\theta_1 + \theta_2) + \cos^2(\theta_1 - \theta_2)} \quad (21)$$

To define the final fitting function $y(\theta_1)$, we incorporate Snell's Law to express θ_2 in terms of θ_1 and add a vertical offset parameter C to account for systematic errors:

$$y(\theta_1) = \left[\frac{\cos^2(\theta_1 + \theta_2)}{\cos^2(\theta_1 + \theta_2) + \cos^2(\theta_1 - \theta_2)} \right] + C \quad (22)$$

where $\theta_2 = \arcsin(\frac{n_1}{n_2} \sin \theta_1)$, from Snell's Law. This yields a model with two free parameters: the refractive index n_2 and the offset C .

2. The total weighted sum of squared residuals (χ^2) was calculated. A weighted non-linear least squares regression was then performed using the Excel Solver tool, which minimized the χ^2 by optimizing the refractive index and the systematic offset. Weights were defined as the inverse square of the uncertainty ($w_i = 1/\sigma_i^2$)⁶. The results were confirmed by the *nls()* (nonlinear least squares) function in R, which was also used to calculate the uncertainty in the optimised parameters.

⁶J.R. Taylor, *An Introduction to Error Analysis*, 2nd ed. Eqs. 7.11–7.12.

3. While the signal fluctuated approximately 0.4% (peak-to-peak), a conservative uncertainty of $\delta R_s = \delta R_p = \pm 0.5\%$ was used. This value accounts for random noise fluctuations, variations in the strength of the laser beam, and inaccuracies of the sensor.
4. For clarity, the notation R_p and R_s is replaced with P and S respectively. The ratio of parallel intensity to total intensity is defined as:

$$y = \frac{P}{P + S} \quad (23)$$

The uncertainty in this ratio, σ_y , is determined by propagating the instrumental uncertainties δP and δS using the general variance formula⁷:

$$\sigma_y = \sqrt{\left(\frac{\partial y}{\partial P} \delta P\right)^2 + \left(\frac{\partial y}{\partial S} \delta S\right)^2} \quad (24)$$

Assuming that $\delta P = \delta S = \delta$, this simplifies to:

$$\sigma_y = \frac{\delta \sqrt{P^2 + S^2}}{(P + S)^2} \quad (25)$$

5. To quantify the goodness of the non-linear fit, the reduced chi-squared statistic (χ_ν^2) was calculated. (See Appendix A)
6. The experimental procedure was repeated for three independent trials to assess the reproducibility of the results, and improve the statistical precision of the derived Brewster's angle.

5.3 Results

The raw data and theoretical fit for all 3 runs are plotted on the next page.

⁷J.R, Taylor. An Introduction to Error Analysis (2nd Edition), Eq 3.47

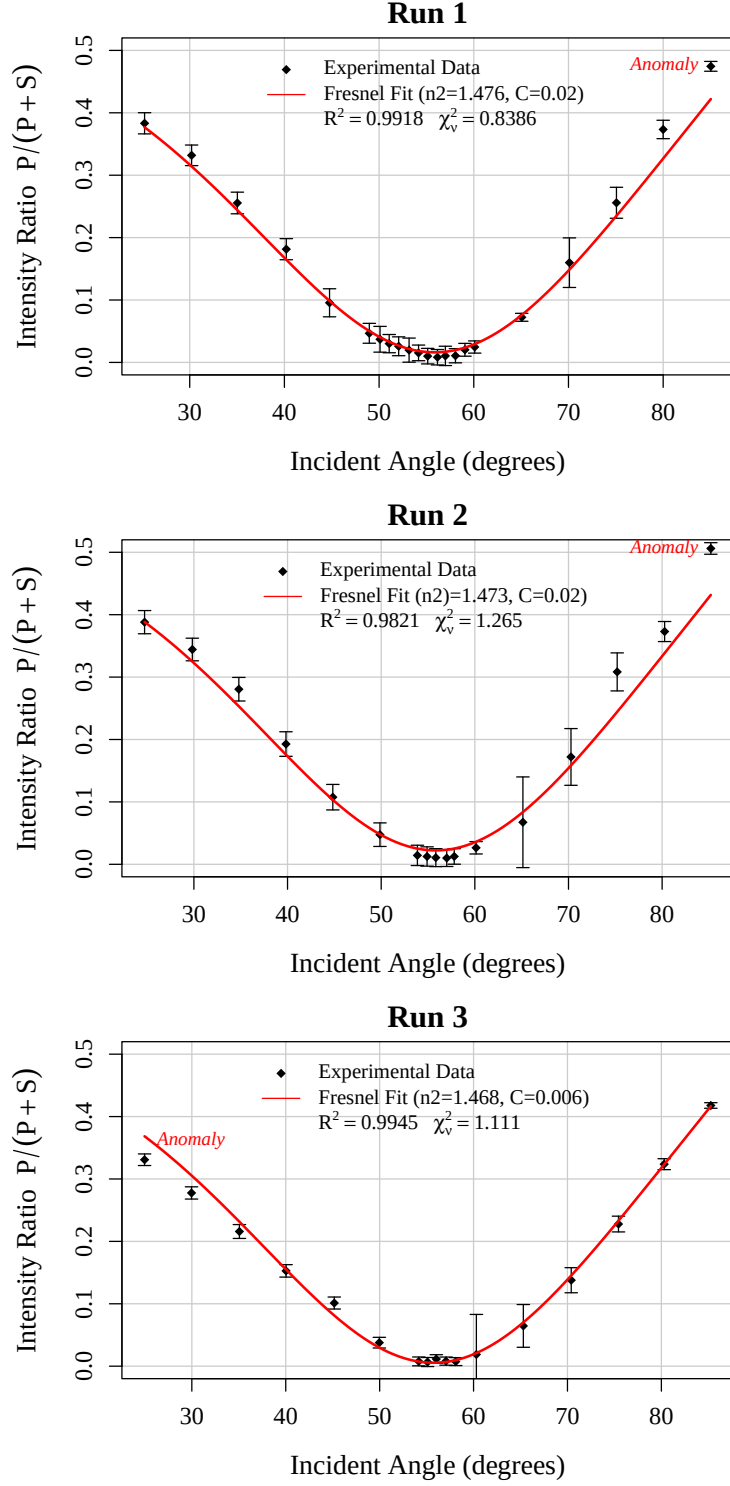


Figure 4: Graph of Intensity Ratio against Incident Angle for all 3 runs

5.3.1 Data Analysis

With high R^2 values and $\chi^2_v \approx 1$, the data was deemed to be statistically consistent.

The data was then combined using a weighted average, with weights defined as the inverse square of the uncertainty ($w_i = 1/\sigma_i^2$). The uncertainty in the final result was determined by $\sigma = 1/\sqrt{\sum w_i}$ ⁸.

$$n_2 = 1.47 \pm 0.01 \quad (26)$$

The value for the vertical offset was also calculated using the same method,

$$C = 0.007 \pm 0.005 \quad (27)$$

and this corresponds to a Brewster's Angle of

$$\theta_{Br} = 55.8 \pm 0.2^\circ \quad (28)$$

5.3.2 Anomalous Points

During the process of calculating the χ_ν^2 values, 3 anomalous points stood out as obvious outliers due to their suspiciously large contribution to the χ_ν^2 values. These data point were then removed from the data set when calculating the parameters of the fitting equation.

Table 1: Statistical Analysis of Experimental Anomalies

Experimental Run	Statistical Deviation	Probability of Occurrence
Run 1 (Angle: 85°)	5.72 σ	1.07×10^{-8}
Run 2 (Angle: 85°)	6.21 σ	5.49×10^{-10}
Run 3 (Angle: 25°)	3.70 σ	2.16×10^{-4}

The possible reasons for these anomalies will be discussed in Section 8: Discussion.

6 Experiment 3: Critical Angle Method

6.1 Procedure

After calibration, the D-lens was rotated such that the laser beam was incident on the curved surface of the lens. The angle of incidence was incrementally increased just until there was no longer any refracted light.

The angle of the lens holder was reset and the critical angle was measured 3 times, from which the average angle was calculated.

6.2 Calculated Values and Error Propagation

The refractive index and error propagation were calculated with the following methods:

1. The refractive index of the acrylic lens was calculated using

$$n_2 = \frac{1}{\sin \theta_c} \quad (29)$$

2. The uncertainty of θ was estimated as half of the smallest interval of the goniometer, plus 1° to account for the ambiguity of when exactly the intensity of refracted light went to zero, added together in quadrature.

$$\delta\theta = \left(\sqrt{(0.5^\circ)^2 + (1.0^\circ)^2} \right) \times \frac{\pi}{180^\circ} \approx 0.0195 \text{ rad} \quad (30)$$

⁸J.R. Taylor, *An Introduction to Error Analysis*, 2nd ed. Eqs. 7.11–7.12.

3. The uncertainty of $\sin \theta$ was then calculated:

$$\sigma_{\sin \theta} = |\cos \theta| \times \delta \theta \quad (31)$$

4. And finally, the uncertainty in n_2 was calculated:

$$\sigma_n = \left| \frac{1}{\sin^2 \theta} \right| \times \sigma_{\sin \theta} \quad (32)$$

6.3 Results

The refractive index of the D-lens, using this method, was found to be

$$n_2 = 1.44 \pm 0.03 \quad (33)$$

which corresponds to a Brewster's Angle of

$$\theta_{Br} = 55.2 \pm 0.5^\circ \quad (34)$$

A detailed comparison with accepted values and analysis of systematic errors is presented in Section 8: Discussion.

7 Global Analysis

A summary of the results from the 3 experiments are shown in the table below.

	Snell's Law	Polarisation Method	Critical Angle Method
Refractive Index	1.40 ± 0.09	1.47 ± 0.01	1.44 ± 0.03
Brewster's Angle	$55 \pm 2^\circ$	$55.8 \pm 0.2^\circ$	$55.2 \pm 0.5^\circ$

Table 2: Summary of Results

As the error bars of the results overlap, they are statistically consistent and can be combined using a weighted average. The weights are given by $w_i = \frac{1}{\sigma_i^2}$, and the weighted uncertainty is $\sigma = \frac{1}{\sqrt{\sum w_i}}$. The final results are:

$$\bar{n} = 1.466 \pm 0.009 \quad (35)$$

which corresponds to a Brewster's Angle of

$$\bar{\theta}_{Br} = 55.7 \pm 0.2^\circ \quad (36)$$

8 Discussion

8.1 Physical Interpretation

The observed intensity minimum at $\theta_{Br} = 55.7^\circ$ provides direct validation of Fresnel's equations and Maxwell's electromagnetic theory. Physically, this corresponds to the condition where the reflected and refracted rays are exactly orthogonal (90° apart), which suppresses the reflection of the p-polarized electric field. Furthermore, the strong agreement between the refractive index derived from this polarization minimum ($n \approx 1.466$) and the critical angle ($n \approx 1.44$) successfully unifies these findings with the predictions of Snell's Law.

8.2 Comparison with Theory

Direct comparison with literature is limited as the true material of the lens is unknown. Although the apparatus manual specifies the material as Acrylic, preliminary tests on the two available D-lenses revealed distinct critical angles of 44° and 42° , corresponding to a refractive index range of $n \approx 1.44 - 1.49$.

Assuming the standard literature value for Acrylic ($n \approx 1.489$), the experimental result ($n = 1.466 \pm 0.009$) represents a discrepancy of 2.6σ . While the result remains physically reasonable (within 1.5%), this indicates a statistically significant deviation from the textbook value, likely due to the uncertainty of the true material of the sample. However, when compared internally against the critical angle method measured for this specific lens ($n \approx 1.44 \pm 0.03$), the results show excellent agreement with a discrepancy of only 0.8σ , confirming the internal consistency of the experimental measurements.

8.3 Error Analysis

8.3.1 Systematic Errors

The most significant systematic error was the centering of the D-lens relative to the rotational axis. Sensitivity checks revealed that a lateral displacement of the lens by only 1 – 2 mm resulted in deviations in the measured refracted angle of a few degrees. Since perfect alignment is unachievable, a small systematic error is likely to occur for this method.

8.3.2 Random Errors

Polarisation Method: The largest source of random uncertainty was fluctuations in the laser intensity. The readings were observed to oscillate predictably, with a magnitude of approximately 0.4% (peak-to-trough).

The second largest source was surface imperfections on the D-lens. There were visible micro-scratches and dust on the surface, which scatter light randomly. This scattering means that the theoretical equations for polarized intensities no longer strictly hold.

The third source of random uncertainty was background ambient light, such as light from computer screens and movement within the room, which introduced noise to the measurements.

Geometrical Methods: For the Snell's Law and Critical Angle experiments, the largest random uncertainty was the resolution of the measuring equipment. The angular position could only be resolved to a precision of $\pm 0.5^\circ$.

8.4 Discrepancies

8.4.1 Experiment 1: Snell's Law

The linear regression analysis yielded a y-intercept of -0.01 . While theory predicts an intercept of exactly zero, this discrepancy is statistically insignificant. The theoretical value falls well within the experimental uncertainty range of the intercept (± 0.05). This deviation is fully accounted for by the random uncertainty in the angular resolution of the measuring equipment.

8.4.2 Experiment 2: Polarisation Method

Vertical Offset: A non-zero vertical offset was observed in the intensity curve. Theoretically, the fitted parameter C should represent only detector noise, but it was found to be consistently positive. This discrepancy is accounted for by background ambient light. Since the laboratory environment was not completely dark, the sensor measured some stray light, introducing a systematic additive error and a secondary random error due to fluctuations in ambient brightness.

Anomalies: Three data points were identified as significant statistical outliers, deviating from the theory by multiple standard deviations. Since the errors did not occur at the same incident angle each time, and because of the exceptionally large deviations, they were likely caused by random surface imperfections (such as dust or micro-scratches) on the D-lens.

8.5 Limitations and Future Improvements

A serious issue in this experiment was that the error bars for certain angles (with low total intensity) was disproportionately larger than the rest, as seen in Figure 3. Taking a look at the math, Eq.25 reveals that the size of the error bars is inversely proportional to the square of total intensity:

$$\sigma_y \propto \frac{1}{(P + S)^2} \quad (37)$$

This suggests that to minimise the uncertainty in the calculated ratio, the intensity of the incident beam should be maximised for each angle before measurement. For this particular setup, this can be achieved by changing the angle of the rotating polariser to allow more light to pass through.

9 Conclusion

The refractive index of the D-lens was successfully characterized using three independent methods, resulting in a final weighted mean of $\bar{n} = 1.466 \pm 0.009$ and a Brewster's angle of $\bar{\theta}_{Br} = 55.7 \pm 0.2^\circ$.

The fact that these different test methods agreed so closely suggests that the experimental results are highly accurate. This indicates that the gap between the measured data and the official "acrylic" value ($n \approx 1.489$) is likely due to the specific material properties of the lens provided, rather than a mistake in the experimental procedure. While fluctuations in laser intensity and difficulties in centering the lens remained the primary challenges, the consistent results across all methodologies confirm the validity of the Fresnel equations.

10 References

1. J.R. Taylor (1997). An Introduction to Error Analysis (2nd ed). University Science Books, Sausalito.

Appendix A Reduced Chi-Squared

The formula used to calculate the Reduced Chi-Squared Statistic χ_ν^2 is shown below. This metric normalizes the sum of squared residuals by the variance of the data points and the degrees of freedom.

$$\chi_\nu^2 = \frac{1}{\nu} \sum_{i=1}^N \frac{(y_i - y(\theta_i))^2}{\sigma_{y,i}^2} \quad (38)$$

where:

- y_i is the experimental intensity ratio at angle θ_i ,
- $y(\theta_i)$ is the theoretical value predicted by the Fresnel model,
- $\sigma_{y,i}$ is the propagated uncertainty for that specific data point,
- $\nu = N - p$ represents the degrees of freedom, with N being the total number of data points and $p = 2$ representing the free parameters (n_2 and C).